Leveraging artificial and human intelligence to understand human reinforcement learning

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IICCSSS
August 18, 2026

Today's Agenda

1. Get prerequisites running in CoLab:

https://colab.research.google.com/drive/1zfeo_L9oW0k5HSLibrdgLirt9hzaXpPT?usp=sharing

2. Understand the dataset we're working with.
3. Quick primer on artificial neural network models of human behavior.
4. Implement ANN models of behavior on the dataset.
5. Interpret!
6. Complement ANN models with more traditional models of RL.
7. Freestyle!

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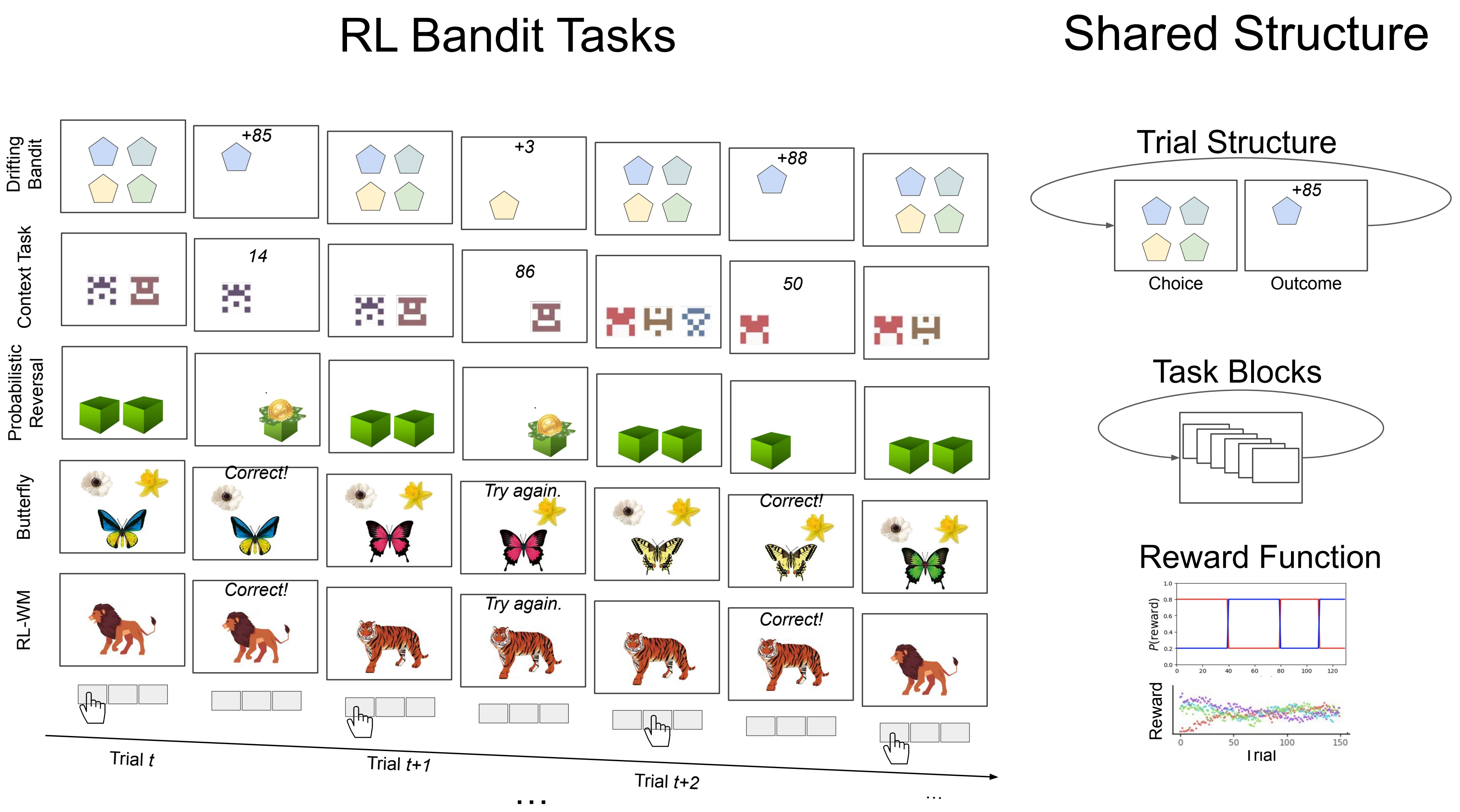
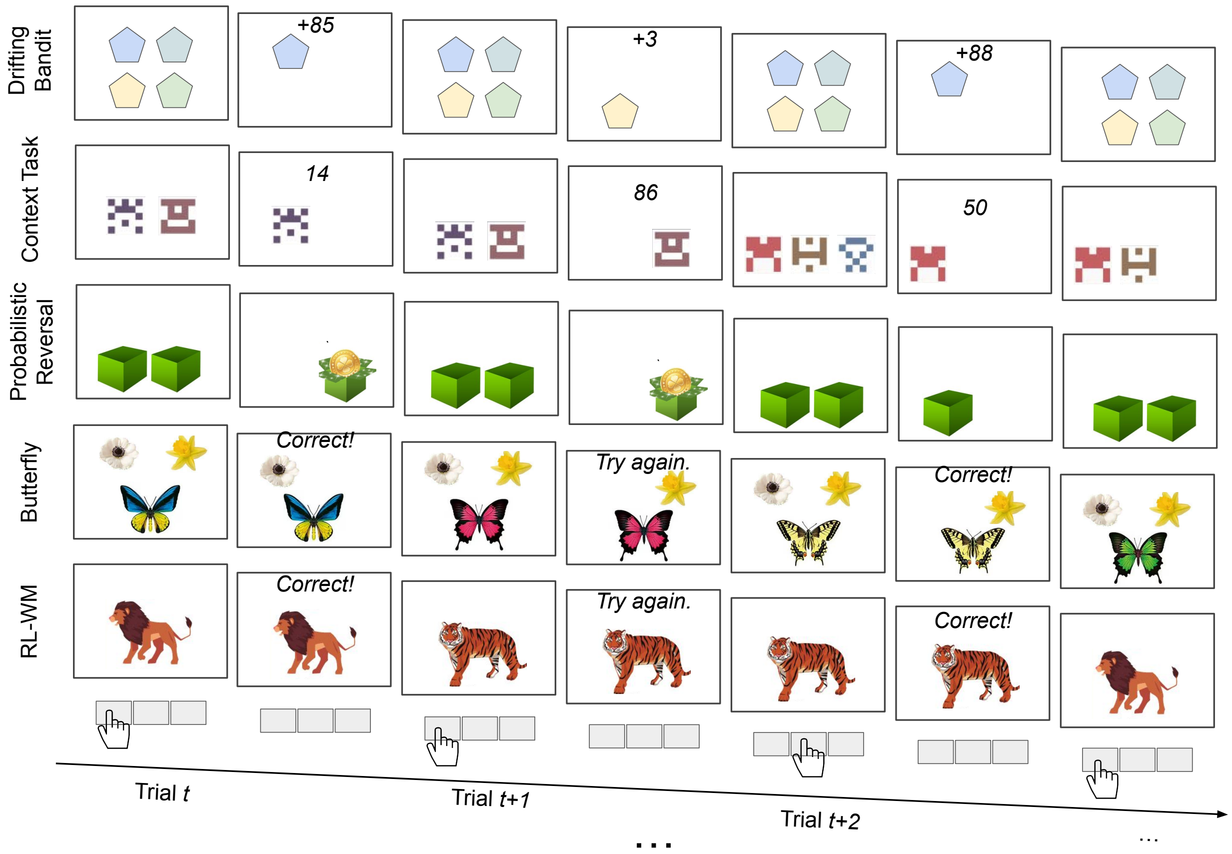


RL Bandit Tasks
Shared Structure
Drifting Bandit
Context Task
Probabilistic Reversal
Butterfly
RL-WM
+85
+3
+88
14
86
50
Correct!
Try again.
Correct!
Correct!
Try again.
Correct!
Trial t
Trial t+1
Trial t+2
...
Trial Structure
+85
Choice
Outcome
Task Blocks
Reward Function
P(reward)
Reward
Trial

RL Bandit Tasks



What have we learned?

A lot!

1. Human learning is impacted by a lot more than what happened 1 trial back. (Eckstein et al. 2025)
2. Memory for value is highly impacted by reward context. (Palminteri et al., 2015)
3. Kids learn slower than adults, adolescents faster! (Master et al., 2020)
4. Working memory plays a key role in human learning. (Collins & Frank, 2012)
5. As does episodic memory (Shohamy & Wagner, 2008)
6. Reward volatility impacts learning speed. (Behrens et al., 2007)

But different tasks yield different findings…

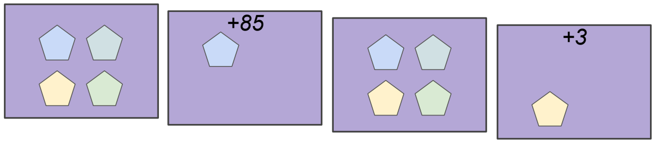
How can we put it all together?

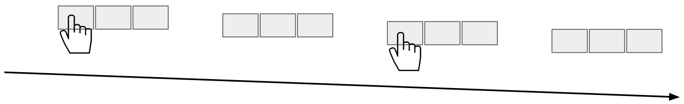
How can we make sense of different findings from different tasks?

What if we moved from tasks to task spaces instead?

Existing RL tasks tile a task space

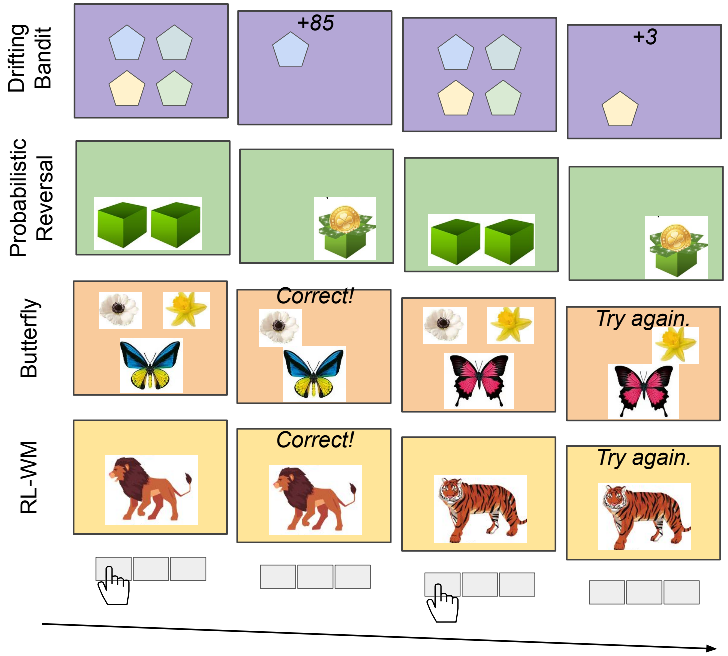
Drifting Bandit





Observability	Actions only	States only	Actions & states	Actions x states		
State space	1	2	3	4	5	6
Action space	2	3	4	5	6	
Reward function (magnitude)	All 1	Continuous & stable	Continuous & drifting			
Reward function (relationship of magnitudes)	Identical	Anti-symmetric	Independent			
Reward function (probability)	All 1	Binary & stable	Continuous & stable	Continuous & drifting		
Reward function (relationship of probabilities)	Identical	Anti-symmetric	Independent			
Volatile feature	States	Rewards type	Probability type			
Volatility "level"	Low	High				

Existing RL tasks tile a task space



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Volatile feature	States	Rewards type	Probability type			
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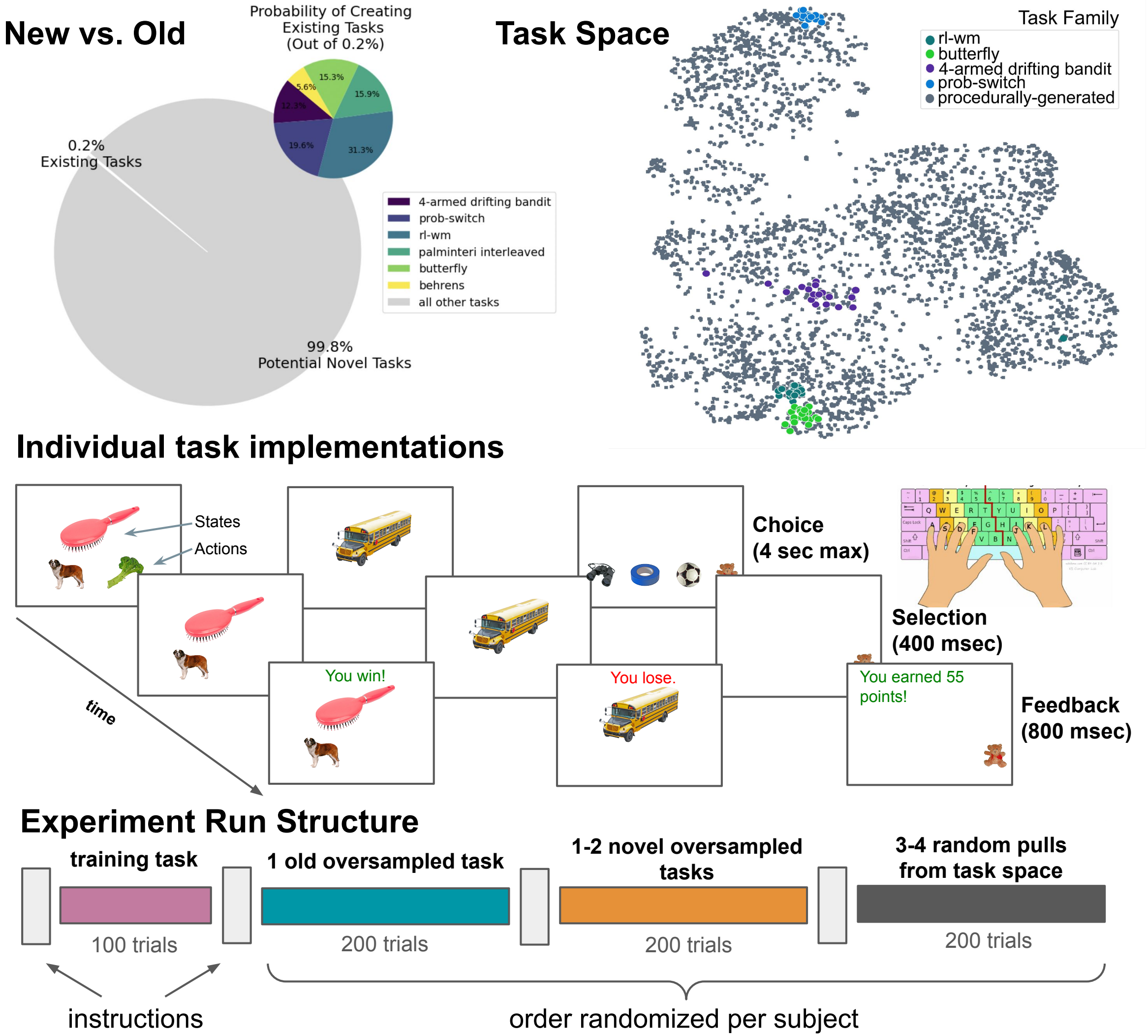
Task 1 (RLWM)
Task 2 (Butterfly)
Task 3 (4-armed drifting bandit)
Task 4 (Probswitch)


New vs. Old
Probability of Creating Existing Tasks (Out of 0.2%)
15.3%
5.6%
15.9%
12.3%
19.6%
31.3%
0.2% Existing Tasks
99.8% Potential Novel Tasks
4-armed drifting bandit
prob-switch
rl-wm
palminteri interleaved
butterfly
behrens
all other tasks
Task Space
Task Family
rl-wm
butterfly
4-armed drifting bandit
prob-switch
procedurally-generated
Individual task implementations
States
Actions
time
You win!
You lose.
Choice (4 sec max)
Selection (400 msec)
You earned 55 points!
Feedback (800 msec)
Experiment Run Structure
training task
100 trials
1 old oversampled task
200 trials
1-2 novel oversampled tasks
200 trials
3-4 random pulls from task space
200 trials
instructions
order randomized per subject

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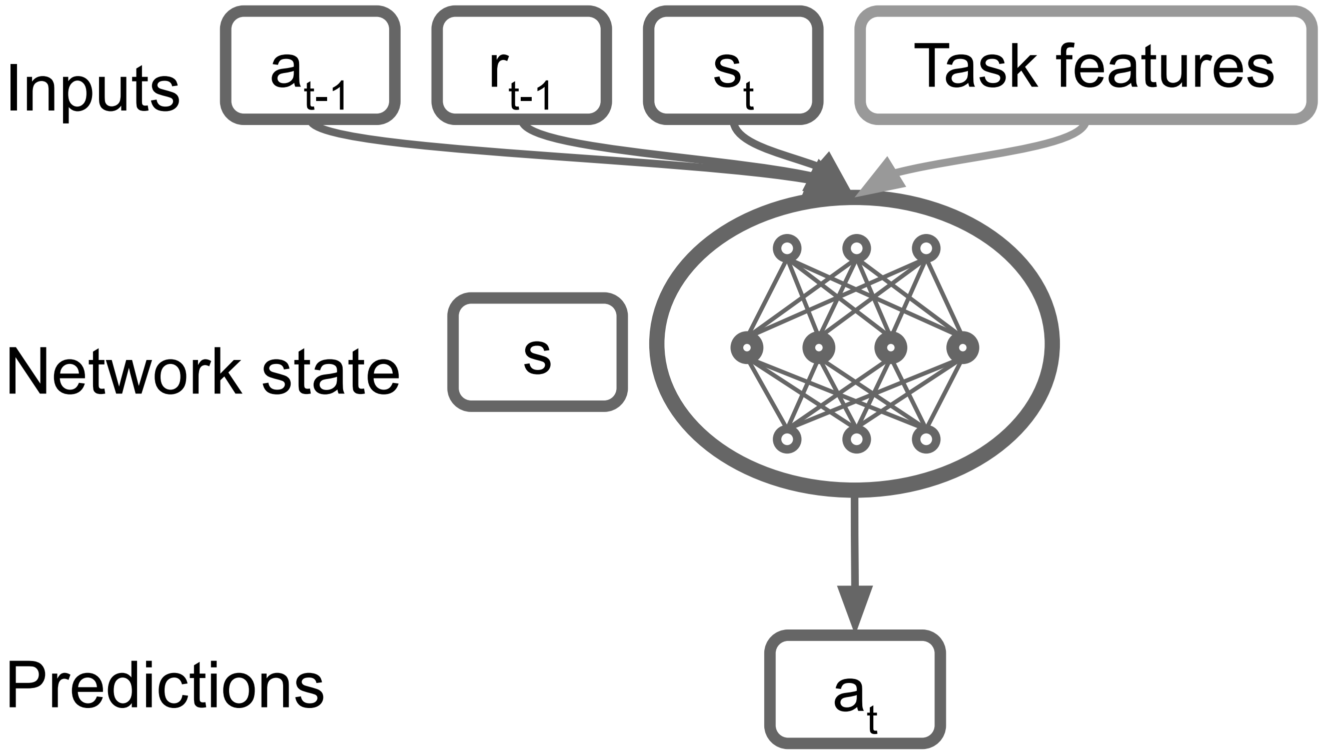
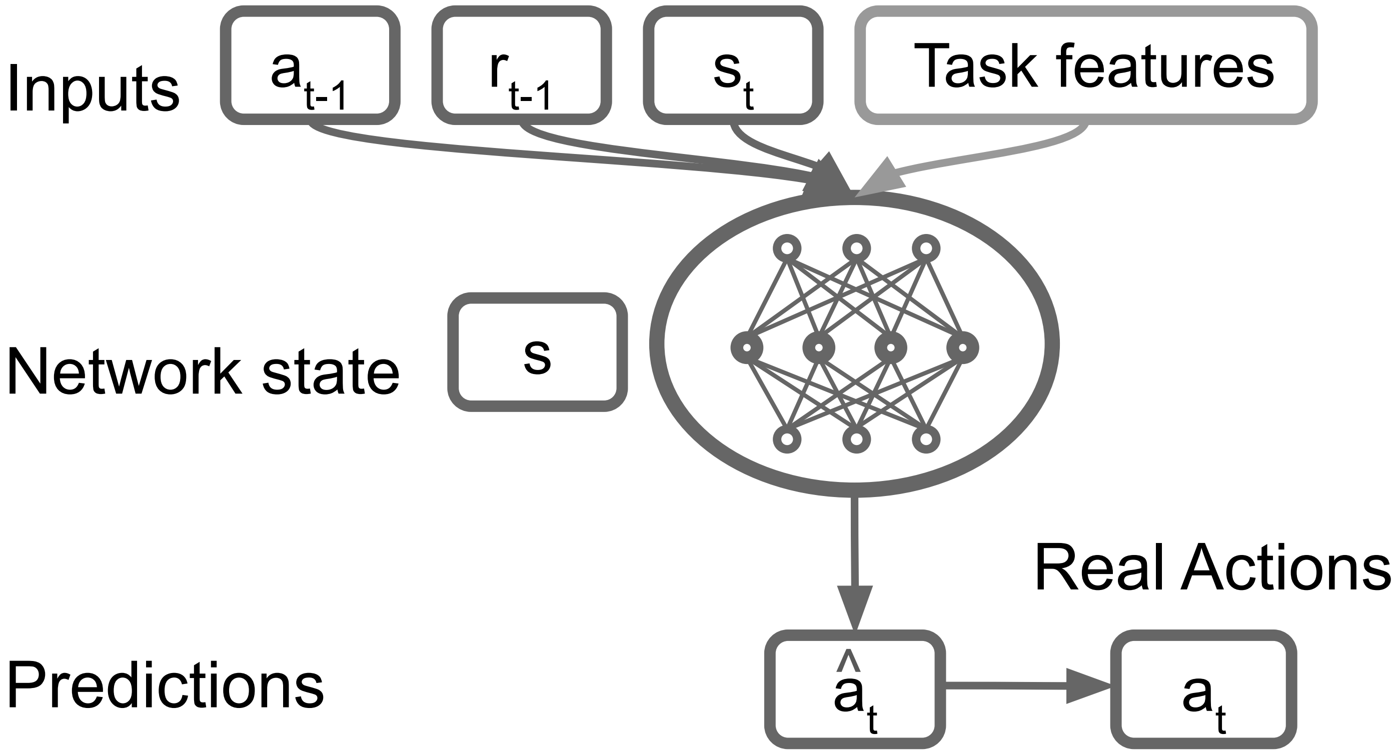
Artificial Neural Networks



Artificial Neural Networks



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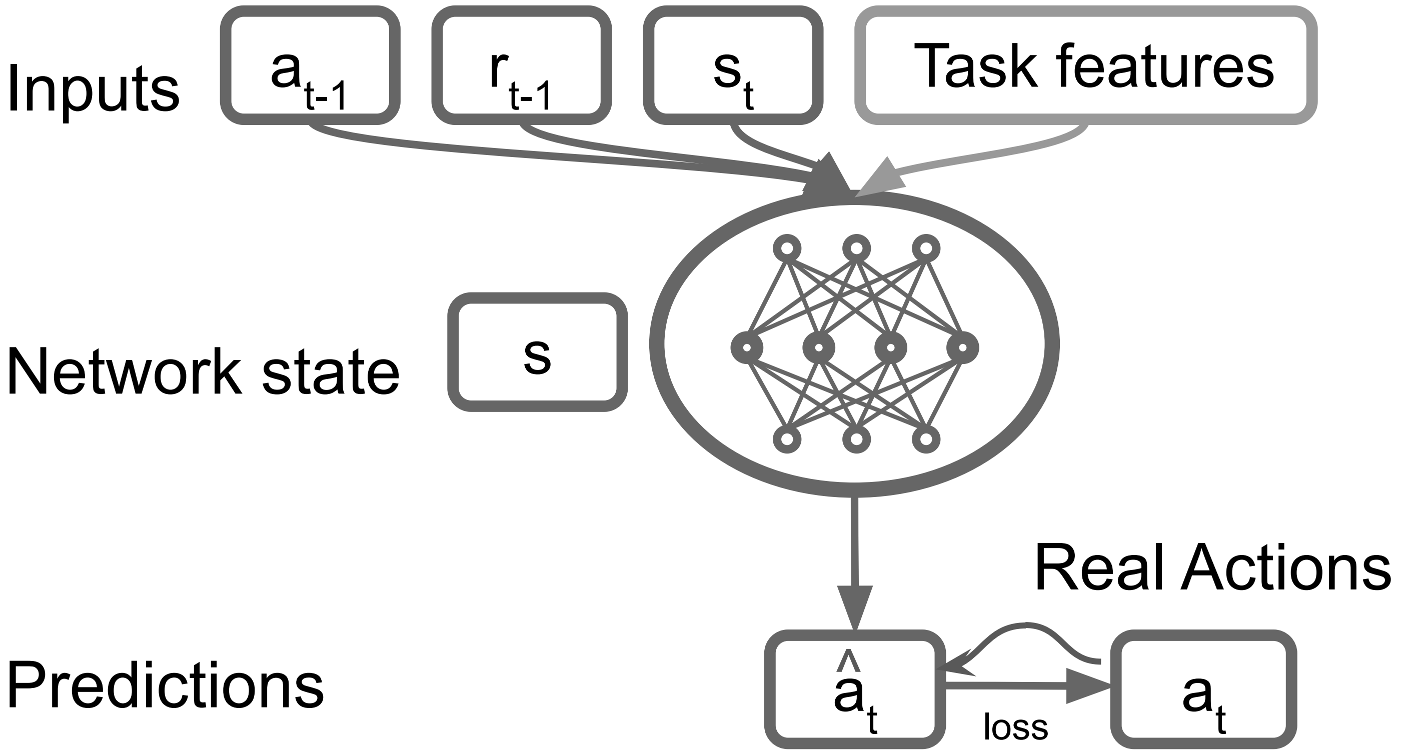
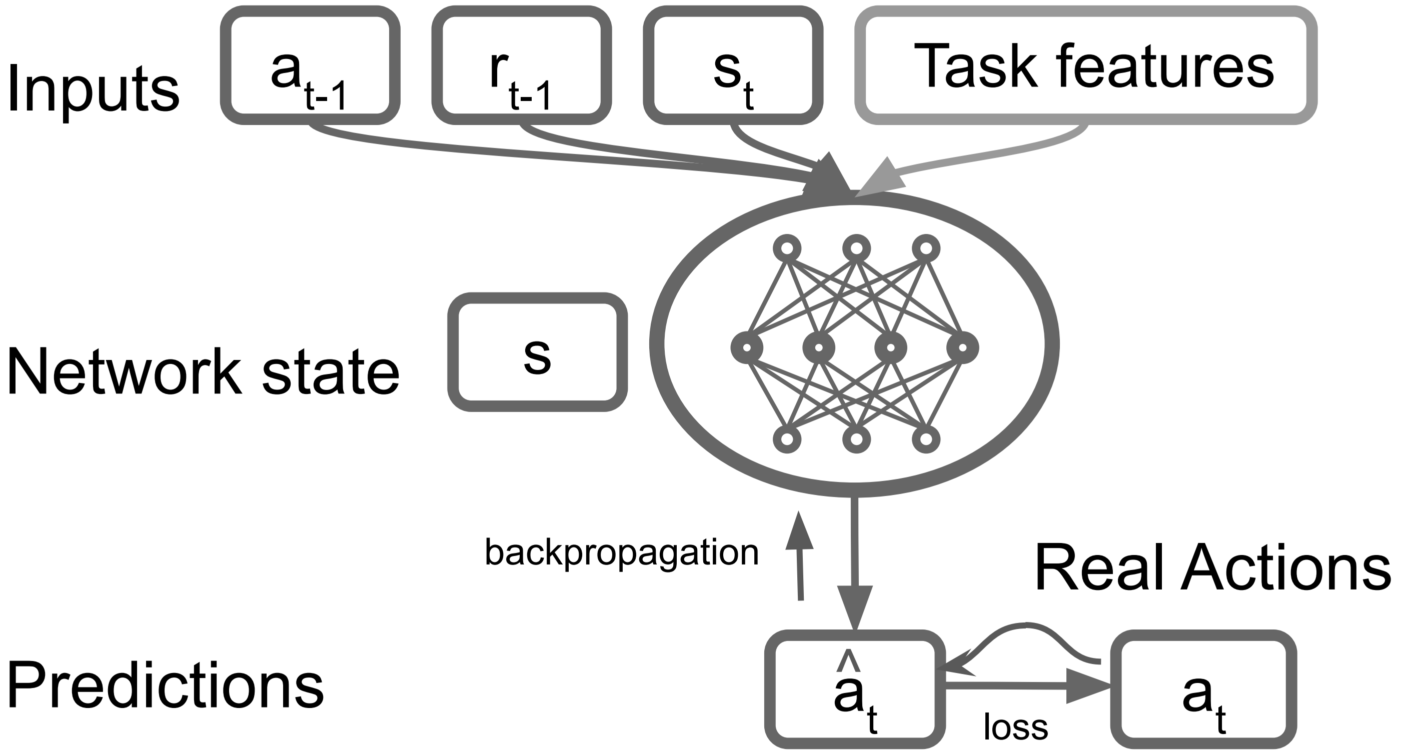
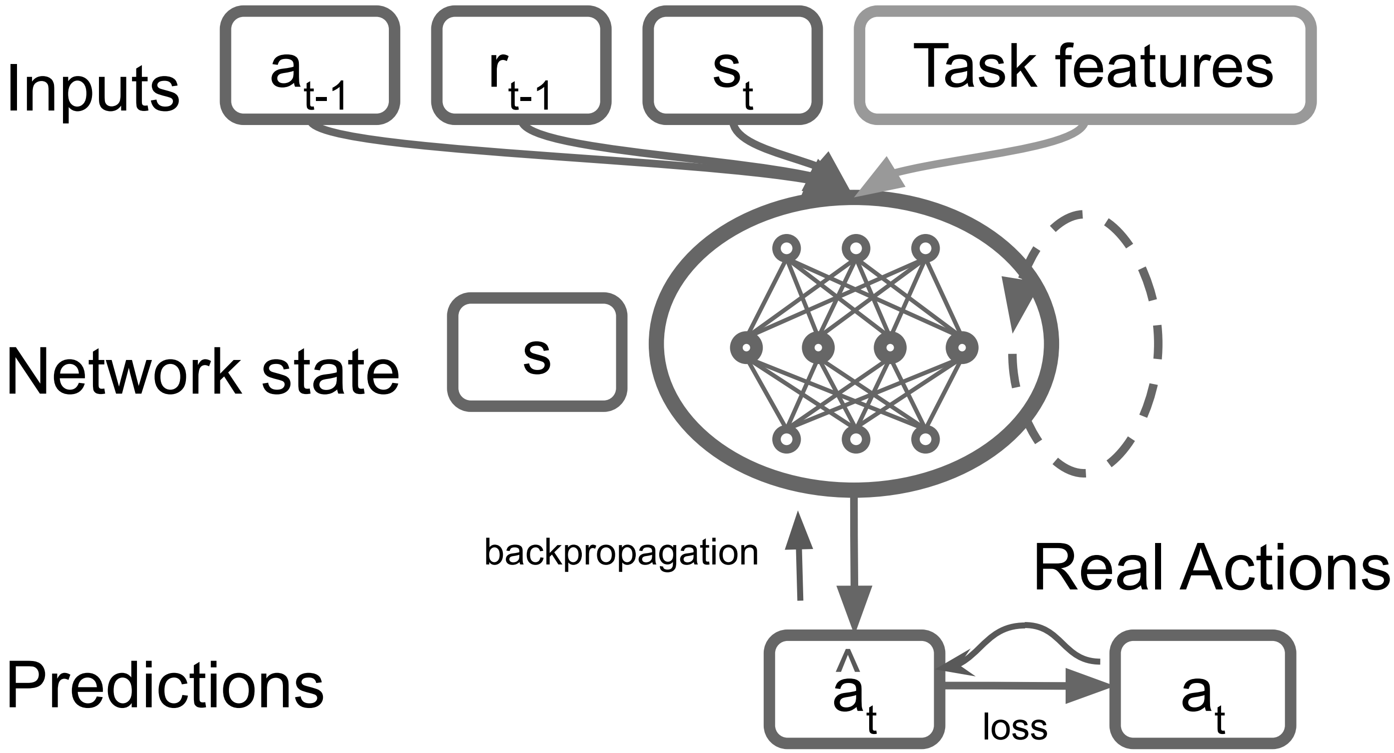


Artificial Neural Networks

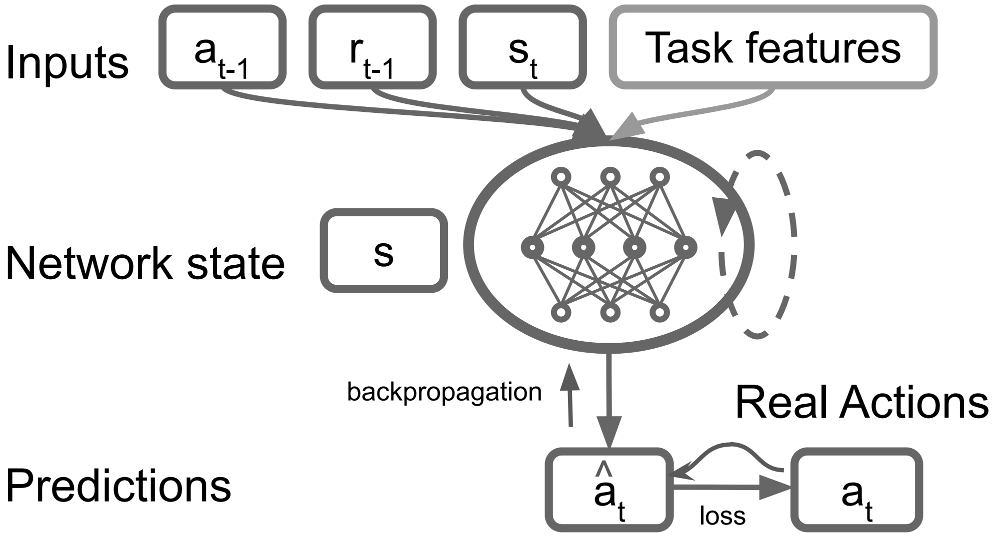


Recurrent Neural Networks



What can these models tell us about human behavior?

1. If it's predictable/regular
2. If it's similar even in diverse areas of task space
3. What areas of task space are most useful for generating novel and/or generalizable predictions



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   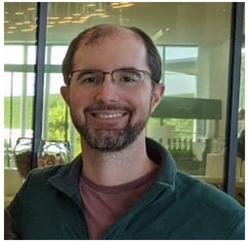 

10,000 tasks for one general theory of human RL

Sarah L. Master, Martin Engelcke, Kevin J. Miller, Kim Stachenfeld, Nathaniel Daw, & Maria K. Eckstein